

RESEARCH

Open Access



Remote clinical decision support tool for Parkinson's disease assessment using a novel approach that combines AI and clinical knowledge

Harel Rom^{1,2}, Ori Peleg^{1,3}, Yovel Rom⁴, Anat Mirelman^{1,2,3}, Gaddi Blumrosen^{5,6} and Inbal Maidan^{1,2,3*}

Abstract

Background Early diagnosis of Parkinson's disease (PD) can assist in designing efficient treatments. Reduced facial expressions are considered a hallmark of PD, making advanced artificial intelligence (AI) image processing a potential non-invasive clinical decision support tool for PD detection. This study aims to determine the sensitivity of image-to-text AI, which matches facial frames recorded in home settings with descriptions of PD facial expressions, in identifying patients with PD.

Methods Facial image of 67 PD patients and 52 healthy-controls (HCs) were collected via standard video recording. Using clinical knowledge, we compiled descriptive sentences detailing facial characteristics associated with PD. The facial images were analyzed with OpenAI's CLIP model to generate probability scores, indicating the likelihood of each image matching the PD-related descriptions. These scores were used in an XGBoost model to identify PD patients based on the total, motor, and facial-expression item of the MDS-UPDRS, a common scale for assessing disease severity.

Results The image-to-text AI technology showed the best results in identifying PD patients based on the facial expression item ($AUC = 0.78 \pm 0.05$), especially for those with 'mild' facial symptoms ($AUC = 0.87 \pm 0.04$). The motor MDS-UPDRS score followed ($AUC = 0.69 \pm 0.05$), while the total MDS-UPDRS score showed the lowest performance in identifying PD patients ($AUC = 0.59 \pm 0.05$). PD matching probabilities between facial images and sentences revealed significant correlations across all MDS-UPDRS components ($r > 0.23$, $p < 0.0001$).

Conclusions Our results demonstrate the feasibility of using advanced AI in a clinical decision support tool for PD diagnosis, suggesting a novel approach for home-based screening to identify PD patients. This method represents a significant innovation, transforming clinical knowledge into practical algorithms that can serve as effective screening tools.

Clinical trial number MOH_2023-04-16_012535

Keywords Parkinson's disease (PD), Facial-expression, Artificial intelligence (AI)

*Correspondence:
Inbal Maidan
inbalm@tlvmc.gov.il

Full list of author information is available at the end of the article



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Background

Parkinson's Disease (PD) is a progressively debilitating neurological disorder that has become increasingly prevalent over the last three decades due to the rise in life expectancy. It is ranked as the second most common neurodegenerative condition [1, 2]. PD is predominantly presented with motor symptoms such as bradykinesia, rigidity, resting tremor, and postural instability [3]. Identifying PD at relatively early stages can improve the effectiveness of treatment plans and assist in the well-being of PD patients [4]. However, diagnosing PD at early stages is challenging, primarily due to non-specific symptoms that may not exclusively indicate PD and can therefore be missed by non-specialist neurologists, causing a significant delay between a subject's initial PD symptoms and the time of diagnosis.

In recent years, considerable efforts have been devoted to identifying biomarkers that could facilitate the early detection of PD and establish objective, available, and reproducible indicators for PD diagnosis using different sensing modalities and machine learning techniques [4, 5]. Among these efforts, the analysis of facial movements has emerged as a promising approach [6]. Studies have shown that PD affects patients' facial muscle movements, leading to reduced facial expression known as hypomimia, involuntary movements (i.e., tics), and muscle weakness or stiffness [7]. This impact on facial muscles can manifest in both voluntary and spontaneous movements, with the latter being particularly affected during cognitive load, which can exacerbate motor symptoms [8, 9]. Recent research has focused on analyzing facial muscle movements as a potential screening test for PD detection. For instance, Pegolo's study demonstrated that analyzing the distances between pairs of geometric facial features can identify PD patients with an accuracy range of 0.88 to 0.89 [6]. However, while promising, the method of marking and measuring distances between different geometric facial features is complex and time-consuming. This limitation highlights the need for more efficient and accessible approaches to facial analysis in PD screening.

Specialists in movement disorders can use verbal descriptions to convey the visual manifestations of hypomimia and bradykinesia in the faces of PD patients, employing phrases such as 'little play of expression' or 'facial stiffness.' Therefore, we suggest incorporating this human knowledge, as reflected in clinical descriptions of PD, to enhance early detection of the disease. Therefore, the main goal of this study was to identify PD patients based on facial expressions and differentiate them from healthy controls by utilizing advanced AI models in natural home environments. We aimed to match images of PD patients with their clinical descriptions using the image-to-text CLIP model. We hypothesized that this

model will effectively distinguish between PD patients and healthy controls. Our analysis compared the predictive accuracy of our model against facial, motor, and total PD scores derived from medical experts using the MDS-UPDRS [10]. We anticipated that our approach will enhance the identification of PD patients and correlate with item 3.2 of the MDS-UPDRS, which specifically relates to facial expressions.

Methods

Participants characteristics

The study included 119 participants aged 45–85 years, comprising 67 PD patients and 52 HCs. Participants were recruited from the Laboratory of Early Markers of Neurodegeneration (LEMON) at the Neurological Institute, Tel Aviv Sourasky Medical Center (TASMC). Patients were included if they were diagnosed with PD based on the MDS criteria and if they were in stages Hoehn and Yahr (H&Y) between 1 and 3. Exclusion criteria for all participants included any other neurological, orthopedic, or psychiatric disorders, significant cognitive impairment (Montreal Cognitive Assessment cut-off score of <21) and Botox injections. The MDS-UPDRS and MOCA were obtained during routine visits to LEMON at the Tel Aviv Sourasky Medical Center. We used data collected in LEMON during visits that occurred within 3 months before or after our task performance, selecting the closest available assessment. All experimental tasks were performed by PD patients in the "ON" state, i.e., approximately 1.5 hours after taking their last medications. The study was approved by the local ethical committee and was performed according to Helsinki guideline [9]. All subjects gave their informed written consent prior to participation.

PD facial clinical descriptions set construction

Leveraging the clinical knowledge accumulated in the literature on impairments in facial expression related to PD [11–14], we documented verbal descriptions of distinct facial expressions and visual characteristics commonly exhibited by PD patients. Based on it, we assembled a comprehensive list of 41 verbal descriptions characterizing facial expressions associated with PD. Let's denote the m 'th description, of the PD score, S_{PD} as $d_m^{S_{PD}}$. The facial description sentences and their corresponding opposites, representing a 0–1 scale where 0 denotes the negative sentence and 1 denotes the positive sentence, are provided in Supplementary A.

Facial videos recording

The research was designed as a remotely conducted paradigm that allowed to examine the subjects at their common, comfortable daily-life environment. In addition, the video was recorded while the subjects performed a

24-minute cognitive task, ensuring the assessment of spontaneous facial movements. During the task, subjects were presented with contextual cues, instructing them to press one of four keys that were visually depicted as assembling a virtual sandwich, and they had to press the cued key on their keyboard. The subjects were instructed and supported by a researcher over the phone and a software package was delivered via email to run the task and record the face video. A computerized remote access monitoring software - Teamviewer® (TeamViewer Inc. 3001 N. Rocky Pt. Dr. E. Ste. 200 Tampa, FL 33607 USA) was used to provide technical support and to ensure that the subjects completed the task correctly. The experiments were mainly conducted in the morning or early afternoon while the patients were approximately 1.5 hours after taking their last medication. Figure 1 below describes the study procedures and analysis.

Facial video processing

In clinical settings, webcam video frames of captured facial expressions include redundancy to save computational power and standardize data lengths while exploring spatial-temporal dynamics. For this, the recorded videos were undersample evenly to obtain $N=1000$ images per session. Let's denote the i 'th subject video clip at the n 'th frame frames as v_i^n , where $n=1...N$. Video frames with partial and absence of the face were excluded.

Image matching with PD clinical scores

An image-to-text AI model (CLIP) [15] was used to evaluate how well each facial image matched the verbal description sentences by generating probability scores. Each sentence was matched to N images sampled evenly

from the video of each participant. For each of these N images, matches were made to each of the 41 verbal descriptions using CLIP.

The matching probability of the n 'th video frame with the m 'th facial description sentence is given by:

$$P_m^{SPD,n} = I(d_m^{SPD}, v_i^n)$$

where I is mutual information measure for the matching probability between the m 'th description, d_m^{SPD} , and the i 'th video frame of subject iv_i^n ,

The probability scores for each verbal description were averaged across the N images, resulting in 41 averaged probability scores per participant. These scores were then used as features to train an XGBoost regression model. The output of the XGBoost model included predictions for the participant's MDS-UPDRS scores: facial expression item (3.2), total motor MDS-UPDRS score (part III), and total MDS-UPDRS score.

Given that our classification was based solely on facial components, we used the specific score given in the facial expression item (3.2) to divide our cohort into the five Likert scale categories: 0 (normal), 1 (slight), 2 (mild), 3 (moderate), and 4 (severe). Since facial expressions also involve motor and cognitive aspects, we wanted to assess how much these additional components, expressed in the total and motor MDS-UPDRS scores, affect our model's ability to classify the patients. Since these two scores encompass various items, unlike the facial item alone, there are no specific cutoffs to define different levels. Therefore, we didn't divide the cohort based on these scores, and calculations for the total and motor

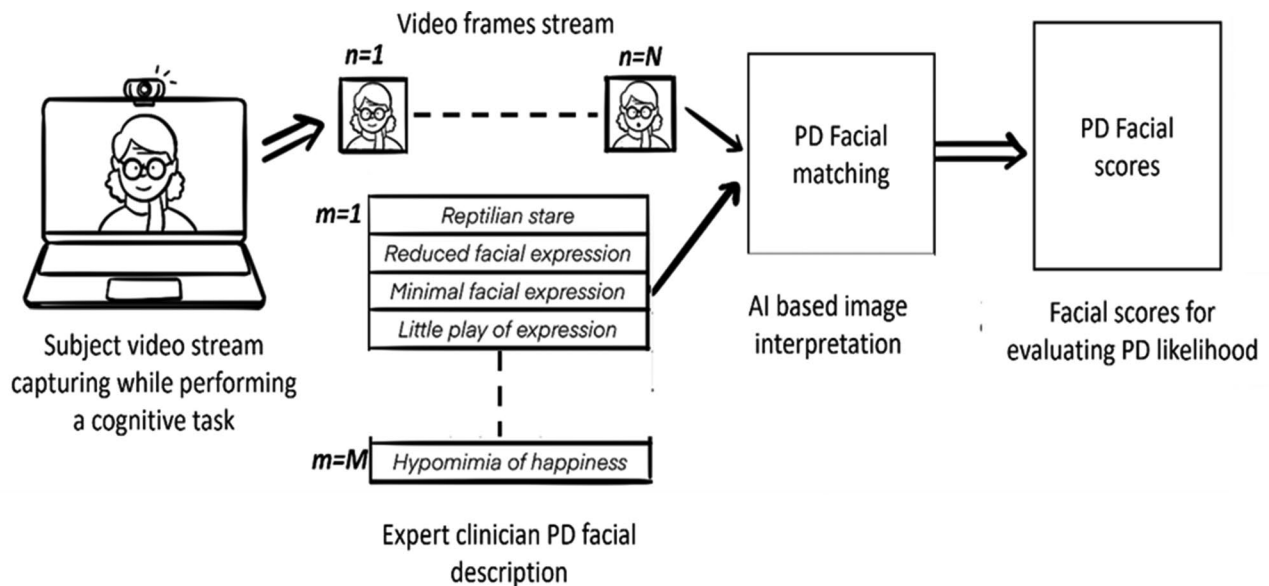


Fig. 1 Deriving PD facial attributes with AI-based matching of the video stream (with N frames) to the set of description sentences (M facial description)

MDS-UPDRS scores were performed for all patients as one group.

Statistical analysis

The XGBoost regression models were trained separately with the target variables being the facial MDS-UPDRS score (item 3.2), the total MDS-UPDRS score, and the motor MDS-UPDRS score (part III). The models included all probability scores, each representing the match between facial images and verbal description sentences. A priori, all verbal description sentences were clinically relevant and derived from the literature on facial expression impairments related to PD [11–14]. Consequently, no variable selection was performed prior to statistical modeling. The regression outputs were fed to a classifier of PD patients or HCs. This classification was tested across all participants using leave-one-out cross validation.

We aimed to evaluate whether the model performs better with patients classified as more severe cases based solely on the facial expression item (3.2). To do this, we examined two classifications: PD patients with ‘slight’ symptoms (1-facial score) versus HCs, and PD patients with ‘mild’ symptoms (2-facial score) versus HCs.

To quantify the classification performance, Area Under the Receiver Operating Characteristic (AUC-ROC) curves were calculated for each MDS-UPDRS score (facial expression item 3.2, motor part, total score). Specifically for the facial expression item 3.2, AUC-ROC curves were calculated separately for the subgroups of PD patients categorized as “slight” and “mild”. In addition, the performance of the regression models was assessed using the correlation coefficient (r), and Spearman correlation to measure the accuracy of the predictions.

Results

Participants

Table 1 below shows that the PD patients and HCs did not differ in age, sex and MOCA. As expected, PD patients had higher MDS-UPDRS total score than HCs.

Based on the facial expression score (MDS-UPDRS item 3.2), as determined by expert assessments using

the MDS-UPDRS scale, 49 patients who received a score of 1 were included in the ‘slight’ patient group, while 18 patients who received a score of 2 were included in the ‘mild’ group.

PD identification via image-to-text AI matching score

We used XGBoost regression models on the matching probabilities to distinguish all PD patients from HCs according to facial expression item 3.2 score (Fig. 2A), total motor MDS-UPDRS score (Fig. 2B), and total MDS-UPDRS score (Fig. 2C). Specifically, for the facial expression item 3.2, we also performed the XGBoost regression model to distinguish PD patients in the “slight” group from HCs, and PD patients in the “mild” group from HCs (Fig. 2D). As shown, the differentiation between all PD patients and HCs was lowest when using the MDS-UPDRS total score ($AUC=0.59\pm0.05$), followed by the total motor MDS-UPDRS score ($AUC=0.69\pm0.05$). The best differentiation was achieved using the facial expression item ($AUC=0.78\pm0.05$), with even higher differentiation between the “mild” group and HCs ($AUC=0.87\pm0.04$).

Correlations between predicted score and clinical scores

Figure 3 illustrates the correlations between the predicted XGBoost regression score (based on matching probabilities between facial images and sentences) and three clinical measures: the facial expression item (MDS-UPDRS item 3.2), the MDS-UPDRS motor part score, and the MDS-UPDRS total score. As expected, our facial-based model showed the strongest correlation with the facial expression score ($r=0.57$, $p<0.001$, $R^2=0.2$; Fig. 3A). The MDS-UPDRS motor part score exhibited a moderate correlation ($r=0.23$, $p<0.01$, $R^2=0.16$; Fig. 3B). The MDS-UPDRS total score showed a slightly weaker correlation ($r=0.24$, $p<0.01$, $R^2=0.08$; Fig. 3C)

Discussion

This study aimed to examine whether facial expressions can be used to identify patients with PD. To achieve our aim, we applied a cutting-edge technique from non-medical domains to annotate images with descriptive sentences, utilizing the CLIP AI model. By leveraging this model, we applied an approach that translates clinical knowledge, expressed as facial description sentences, into algorithms. To our knowledge, this specific method of combining natural language descriptions with image analysis has been rarely explored in the medical field, particularly in the context of PD assessment. Our findings revealed two main points: (1) The AI model for image-to-text matching can identify PD patients with ‘mild’ facial expression severity with an AUC of 0.87 based solely on facial expression, demonstrating similar results to other approaches using specific facial points. (2) Our

Table 1 Participants characteristics

	PD (n=67)	HC (n=52)	PD/HC p-value
Gender (F/M)	30/37	28/24	0.326
Age (years)	66.5±7.7	65.8±9.2	0.682
MOCA	25.7±2.8	26.9±2.5	0.126
MDS- UPDRS (total)	22.9±13.9	1.6±2.3	<0.001
Disease duration (years)	4.8±3.9	na	na
LEDD (mg)	423.1±420.3	na	na

PD=Parkinson’s disease, HC=healthy controls, MOCA=Montreal cognitive assessment, MDS-UPDRS=unified Parkinson’s disease rating scale, LEDD = Levodopa equivalent daily dose. The values represent mean±standard deviation

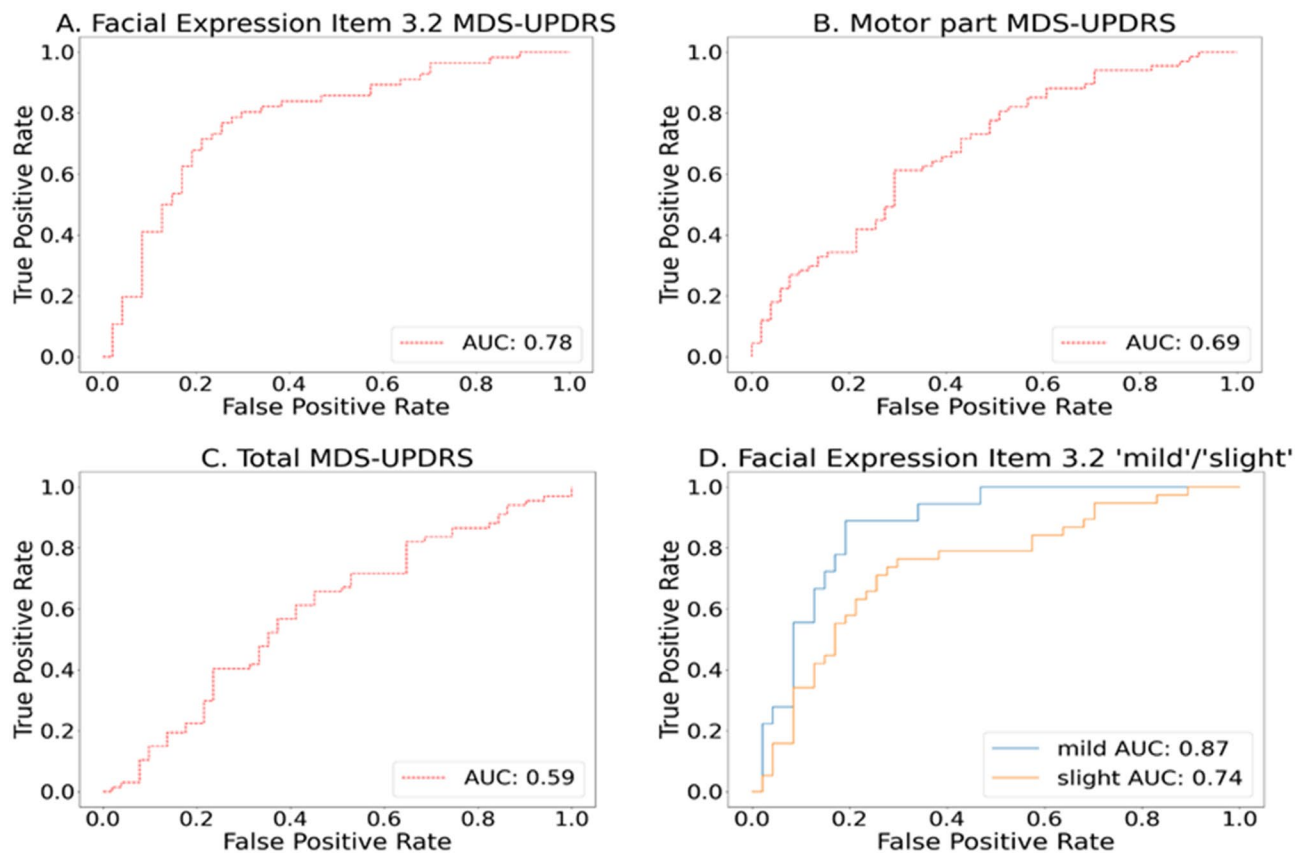


Fig. 2 ROC curves for PD patient classification. **(A)** Classification based on facial expression item 3.2 of MDS-UPDRS. **(B)** Classification using motor MDS-UPDRS (part III) scores. **(C)** Classification using total MDS-UPDRS scores. **(D)** ROC curves for PD patients in the 'slight' group and 'mild' group, based on facial expression item 3.2 scores. AUC values are provided for each curve, indicating the model's discriminative ability

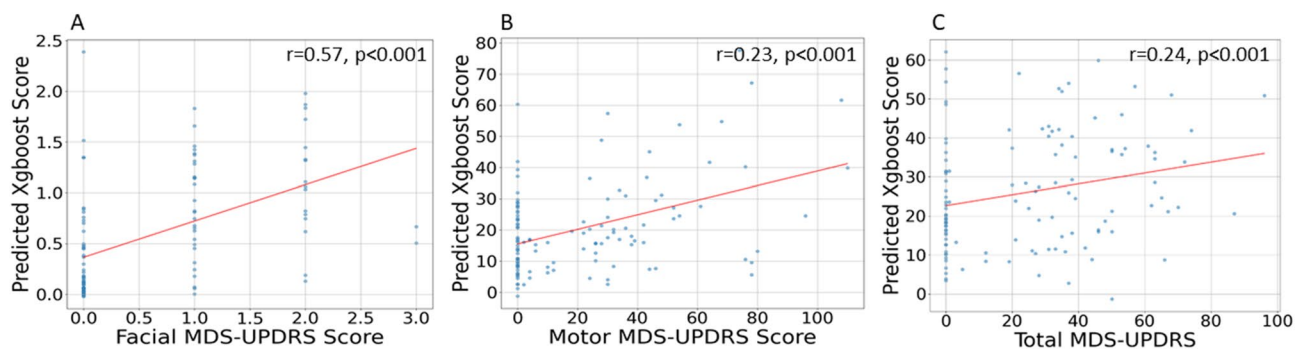


Fig. 3 Spearman correlations between the predicted Xgboost score and **(A)** facial expression (item 3.2), **(B)** the total motor MDS-UPDRS score (part III), and **(C)** the total MDS-UPDRS score across all participants

approach, which focused specifically on facial descriptions, is related to the specific item of facial expression in the MDS-UPDRS but not to other total scores such as the MDS-UPDRS motor score and MDS-UPDRS total score that include additional symptoms of the disease.

Although several studies have reported a reduction in facial movements in patients with PD, it remains unclear whether voluntary, spontaneous, or both types of mimicry are specifically impacted [6, 16, 17]. Our

study focused on spontaneous expressions during cognitive tasks, demonstrating good identification results for PD patients at the mild stage of disease, though not at the slight stage, when compared to recent studies [6, 11, 16–18]. Previous research differs from our study in two main aspects: it relies on voluntary expressions [11, 17, 18], such as explicit 'smile' commands, and it typically involves patients at later stages of the disease [6, 16, 17]. Given that the specific impact on voluntary or

spontaneous mimicry remains unclear, we suggest that integrating both spontaneous and voluntary expressions could be beneficial for improving early-stage detection. The successful identification of PD at early stages is particularly challenging; however, if achieved, it would likely translate to even better generalizability at later stages when disease symptoms become more pronounced.

In addition, our approach is simple and easy to implement in a home setting, compared to the commonly used method of marking various facial points and calculating changes between them, which requires calibration during data collection and signal preprocessing. Our approach has the potential to enhance the field of screening tests for PD. By utilizing accessible technology that can be applied in a home setting without the need for medical professionals, it improves efficiency and accessibility for a broader population of patients and paves the way for earlier identification and treatment. Furthermore, this method could be implemented in other medical fields that involve facial changes along the disease course, potentially enhancing the accuracy and accessibility of medical care for a wide range of conditions.

Current measures of PD motor and non-motor symptoms rely on subjective scales that are infrequently collected and typically administered in clinical settings. The MDS-UPDRS is widely considered the gold standard for assessing Parkinson's disease severity and progression, and likely serves as an important reference for developing new technologies to identify and evaluate the disease [19, 20]. Consequently, it would be expected to correlate well with other measures generated by advanced digital technologies, such as in our study that matches images with clinical knowledge represented by descriptive sentences. However, much of the data captured by such advanced technologies is under-represented or not well assessed by the MDS-UPDRS [21]. This point is reflected in our results showing low AUC when using the total and motor MDS-UPDRS scores. When comparing our results to the specific item related to facial expression, we found the highest AUC, similar to those presented in other studies that used facial methods. Our results demonstrate the specificity of our approach to facial expression, which can be overlooked when compared to non-specific or general measures like the total MDS-UPDRS score.

The goal of new measures is not to replicate what already exists but to assess new outcomes or current ones in advantageous ways (e.g., objectively, frequently, or in real-world settings). Our findings highlight its potential for future studies involving other sub-scores related to various body measurements [22, 23]. Given that the facial sub-score alone has limited quantization, incorporating additional body parts could potentially yield results that surpass those found to date [24, 25]. In general, it would be valuable to combine assessments of multiple body

parts affected by PD. This approach could provide a more comprehensive understanding of the disease progression and potentially reveal correlations between symptoms in different body parts.

There are several limitations to this study that should be noted. Firstly, although dynamic facial expressions were utilized, specific frames were extracted that were static in nature. Secondly, the cognitive task employed was monotonic, and it is possible that the accuracy of the algorithm could be improved by including video footage of subjects engaged in an emotional task. Thirdly, all subjects were in the ON state, approximately 1.5 hours after taking their dopaminergic medications, and therefore the medication may have impacted the results of the study. It is important to note that dopaminergic medications affect motor symptoms related to bradykinesia, tremor, and rigidity, and therefore may influence test results. This should be taken into consideration when performing and interpreting such tests. Future studies may benefit from comparing performance in both 'ON' and 'OFF' states to further elucidate the impact of medication timing on test results.

Conclusions

In this study, we utilized an image-to-text matching AI model to annotate facial images with clinical descriptive sentences. Our findings indicated stronger correlations with the PD facial score, suggesting that focusing on specific MDS-UPDRS components enhances accuracy. To further improve the utility of our approach as a potential screening tool, future work will involve expanding the analysis to include additional body parts, conducting longitudinal studies to track changes over time, increasing the subject population size, and incorporating full-body video recordings in home settings. These efforts aim to enhance early detection of PD symptoms and improve patient outcomes through a more comprehensive understanding of motor and non-motor manifestations of the disease.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12911-025-03104-6>.

Supplementary Material 1

Acknowledgements

Not applicable.

Author contributions

HR analyzed the data, interpreted the results, and was a major contributor in writing the manuscript. OP developed the cognitive task and collected all data. YR was involved in study design and data analysis. AM helped in study design and interpretation of the results. GB was involved in study design, data analysis, interpretation of data, and writing. IM designed the study, interpreted the data, and was a major contributor in writing the manuscript.

Funding

The study was internally funded. The funder played no role in study design, data collection, analysis and interpretation of data, or the writing of this manuscript.

Data availability

The datasets generated and analyzed during the current study are available in GitHub repository, <https://github.com/harel-coffee/Lemon-PD-Faces/blob/master/data.csv>. The underlying code for this study is available in GitHub and can be accessed via this link <https://github.com/harel-coffee/Lemon-PD-Faces/blob/master/Lemon-PD-Faces.ipynb>.

Declarations

Ethics approval and consent to participate

The study was approved by the local ethical committee of Tel Aviv Sourasky Medical Center (TLVMC), and was performed according to the principles of the Declaration of Helsinki. All participants gave their informed written consent prior to participation (Clinical trial number: MOH_2023-04-16_012535).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹Laboratory of Early Markers of Neurodegeneration, Neurological Institute, Tel Aviv Sourasky Medical Center, 6 Weizmann Street, Tel Aviv 64239, Israel

²Sagol School of Neuroscience, Tel Aviv University, Tel Aviv, Israel

³Gray Faculty of Medical and Health Sciences, Tel Aviv University, Tel-Aviv, Israel

⁴AEYE Health, Tel Aviv, Israel

⁵Faculty of Digital Medical Technologies, Holon Institute of Technology (HIT), Holon, Israel

⁶School of Computer Science, Holon Institute of Technology (HIT), Holon, Israel

Received: 24 September 2024 / Accepted: 4 July 2025

Published online: 07 August 2025

References

1. GBD Neurology collaborators. global, regional, and national burden of neurological disorders, 1990–2016: a systematic analysis for the global burden of disease study 2016. *Lancet Neurol.* 2016;(18):459–80.2019
2. Bloem BR, Okun MS, Klein C. Parkinson's disease. *Lancet.* 2021;397:2284–303.
3. Armstrong MJ, Okun MS. Diagnosis and Treatment of Parkinson Disease: a Review. *Jama.* 2020;323:548–60.
4. Gupta R, Kumari S, Senapati A, Ambasta RK, Kumar P. New era of artificial intelligence and machine learning-based detection, diagnosis, and therapeutics in Parkinson's disease. *Ageing Res Rev.* 2023;90:102013.
5. Lotankar S, Prabhavalkar KS, Bhatt LK. Biomarkers for Parkinson's Disease: recent Advancement. *Neurosci Bull.* 2017;33:585–97.
6. Pegolo E, Volpe D, Cucca A, Ricciardi L, Sawacha Z. Quantitative Evaluation of Hypomimia in Parkinson's Disease: a Face Tracking Approach. *Sensors (Basel).* 2022;22:1358.
7. Abrami A, et al. Automated Computer Vision Assessment of Hypomimia in Parkinson Disease: proof-of-Principle Pilot Study. *J Med Internet Res.* 2021;23:e21037.
8. Dirks MF, et al. Cognitive load amplifies Parkinson's tremor through excitatory network influences onto the thalamus. *Brain.* 2020;143:1498–511.
9. Association GAOTWM. World Medical Association Declaration of Helsinki: ethical principles for medical research involving human subjects. *J Am Coll Dent.* 2014;81:14–18.
10. Goetz CG, et al. Movement Disorder Society-sponsored revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS): scale presentation and clinimetric testing results. *Mov Disord.* 2008;23:2129–70.
11. Marsili L, et al. Bradykinesia of posed smiling and voluntary movement of the lower face in Parkinson's disease. *Parkinsonism Relat Disord.* 2014;20:370–75.
12. Maycas-Cepeda T, et al. Hypomimia in Parkinson's Disease: what Is It Telling Us? *Front Neurol.* 2021;11:603582.
13. Ozekmekçi S, Benbir G, Ozdogan FY, Ertan S, Kiziltan ME. Hemihypomimia, a rare persistent sign in Parkinson's disease: follow up of 11 patients. *J Neurol.* 2007;254:347–50.
14. Zingler VC, Strupp M, Jahn K, Brandt T. Hemihypomimia in Parkinson's disease. *Eur Neurol.* 2005;53:92–93.
15. Agarwal S, et al. Evaluating CLIP: towards Characterization of Broader Capabilities and Downstream Implications. Preprint at <http://arxiv.org/abs/2108.02818> (2021).
16. Kang J, Derva D, Kwon D-Y, Wallraven C. Voluntary and spontaneous facial mimicry toward other's emotional expression in patients with Parkinson's disease. *PLoS One.* 2019;14:e0214957.
17. Mattavelli G, et al. Facial expressions recognition and discrimination in Parkinson's disease. *J Neuropsychol.* 2021;15:46–68.
18. Bowers D, et al. Faces of emotion in Parkinson's disease: micro-expressivity and bradykinesia during voluntary facial expressions. *J Int Neuropsychol Soc.* 2006;12:765–73.
19. Hasan H, Athauda DS, Foltynie T, Noyce AJ. Technologies assessing limb bradykinesia in Parkinson's disease. *J Parkinsons Dis.* 2017;7:65–77.
20. Brooks C, et al. Quantification of discrete behavioral components of the MDS-UPDRS. *J Clin Neurosci.* 2019;61:174–79.
21. Kohat AK, et al. Stability of MDS-UPDRS Motor Subtypes Over Three Years in Early Parkinson's Disease. *Front Neurol.* 2021;12:704906.
22. Blumrosen G, Miron Y, Intrator N, Plotnik M. A Real-Time Kinect Signature-Based Patient Home Monitoring System. *Sensors (Basel).* 2016;16:1965.
23. Lavi Y, Birnbaum D, Shabat O, Blumrosen G. Biometric system based on kinect skeletal, facial and vocal features. 2019;880:884–903.
24. Sarapata G, et al. Video-Based Activity Recognition for Automated Motor Assessment of Parkinson's Disease. *IEEE J Biomed Health Inform.* 2023;27:5032–41.
25. Radford A, et al. Learning transferable visual models from natural language supervision. in *International conference on machine learning* 8748–63 (PMLR, 2021).

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.